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LIBRARY MANAGEMENT SYSTEM [LMS]

- PROBLEM STATEMENT NUMBER : 02
- PROBLEM STATEMENT NAME : LIBRARY MANAGEMENT SYSTEM [LMS]
- TEAM NAME : REALISTIC
- TEAM ID : STH24A02



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LibraTech: Smart Library Management System



Proposed Solution:

A modern Library Management System (LMS) designed to streamline library operations. The system ensures efficient management of book inventory, user registrations, book lending/return processes, payment handling, and deadline reminders.

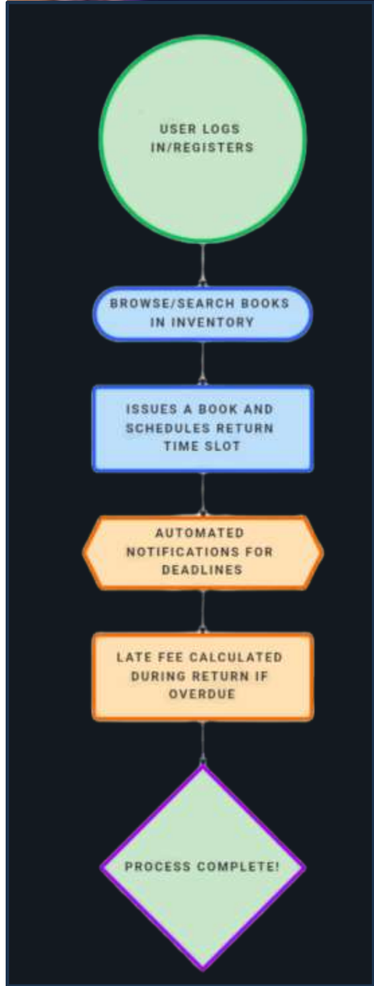
Explanation:

The LMS simplifies complex library workflows for librarians and users alike, incorporating features such as:

- ✓ Book inventory management.
- ✓ Time slot scheduling for borrowing/returning books.
- ✓ Automated notifications for due dates and payments.

Innovation & Uniqueness:

- ✓ Use of AI tools & technologies for recommendations and search optimization (e.g., Google Books API, Elasticsearch).
- ✓ Integration of FullCalendar.js for visual time slot scheduling and management enhanced by Scikit-learn to analyse historical data to predict peak times.
- ✓ Automated reminders via Twilio & SendGrid for SMS and email notifications.
- ✓ Real time book availability tracking with IoT integration (RFID sensors)



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TECHNICAL APPROACH



Methodology and Process:

1. Book Inventory Management
- ✓ CRUD operations for books using MySQL and React.js.
- ✓ Integration with Google Books API for book data.

2. User Management
- ✓ JWT-based authentication using Node.js.
- ✓ Membership fee and borrowing limit management.

3. Book Lending and Return System
- ✓ Late fee calculation using Moment.js.
- ✓ Real-time inventory updates.

4. Time Slot Management
- ✓ Visual scheduling using FullCalendar.js.

5. Deadline Reminders
- ✓ Notifications via Twilio/SendGrid API.

Technologies Used:

1. Frontend
- ✓ React.js
- ✓ Material-UI
- ✓ FullCalendar.js

2. Backend
- ✓ Node.js
- ✓ Express.js
- ✓ MySQL
- ✓ Elasticsearch
- ✓ RFID sensors

3. APIs
- ✓ Google Books API
- ✓ Twilio API
- ✓ SendGrid API

Tech Stack:



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FEASIBILITY & VIABILITY



Feasibility Analysis:

1. TECHNICAL
   - Uses widely adopted frameworks and APIs.

2. OPERATIONAL
   - Designed to handle common library workflows effectively.

3. SCALABILITY
   - Modular design allows future expansion.

Challenges & Risks :

- Integration complexities between frontend and backend.
- Handling real-time updates in inventory and scheduling.

Strategies for Overcoming Challenges:

✓ Use well-documented libraries and frameworks (e.g., React, FullCalendar.js).

✓ Ensure continuous testing with tools like Postman for API debugging.



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IMPACTS & BENEFITS

Benefits:

1. SOCIAL
- ✓ Enhanced Accessibility.
- ✓ User Engagement.
- ✓ Education Growth.

2. ECONOMIC
- ✓ Cost effective.
- ✓ Revenue streams.
- ✓ Scalable design.

3. ENVIRONMENTAL
- ✓ Paperless operations.
- ✓ Resource Optimization.
- ✓ Lower Emissions.

Impacts:

1. OPERATIONAL EFFICIENCY
- ✓ Efficient Workflows.
- ✓ Optimized Staff Roles.

2. EMPOWERED LEARNING
- ✓ Better access to resources.
- ✓ Encourages Reading.

3. ENVIRONMENTAL IMPACT
- ✓ Reduces paper usage.
- ✓ Resource Efficiency.

4. INCLUSIVE SERVICE
- ✓ Broad accessibility.
- ✓ Flexibility.



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REFERENCES

1. Google Books API
   - https://developers.google.com/books

2. FullCalendar.js
   - https://fullcalendar.io

3. Twilio API
   - https://twilio.com

4. SendGrid API
   - https://sendgrid.com

5. Node.js
   - https://nodejs.org

6. React.js
   - https://reactjs.org

7. PostgreSQL
   - https://postgresql.com

8. Machine Learning for personalized recommendations
   - https://www.tensorflow.org/js
   - https://scikit-learn.org

9. IoT Integration
   - https://aws.amazon.com
   - https://azure.microsoft.com

